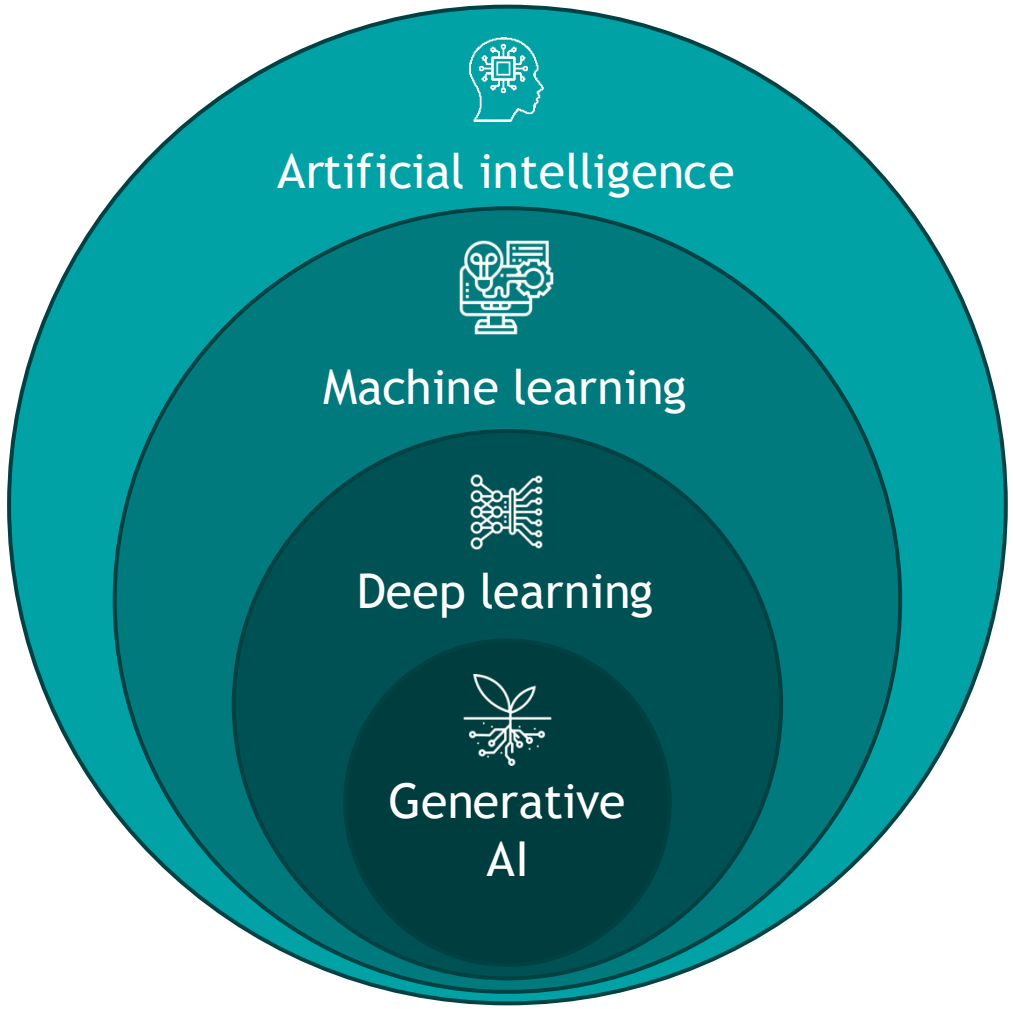
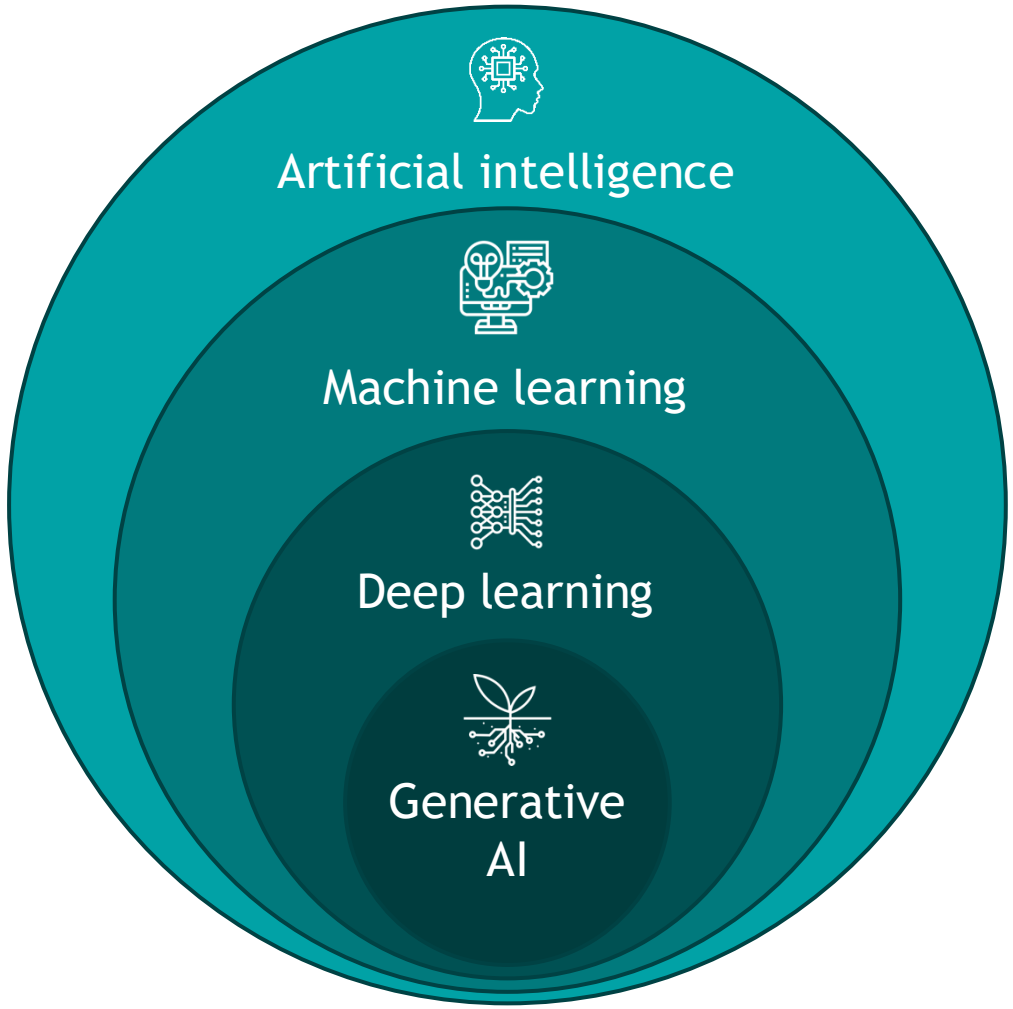
01 Introduction to Machine Learning

Artificial intelligence



Artificial intelligence

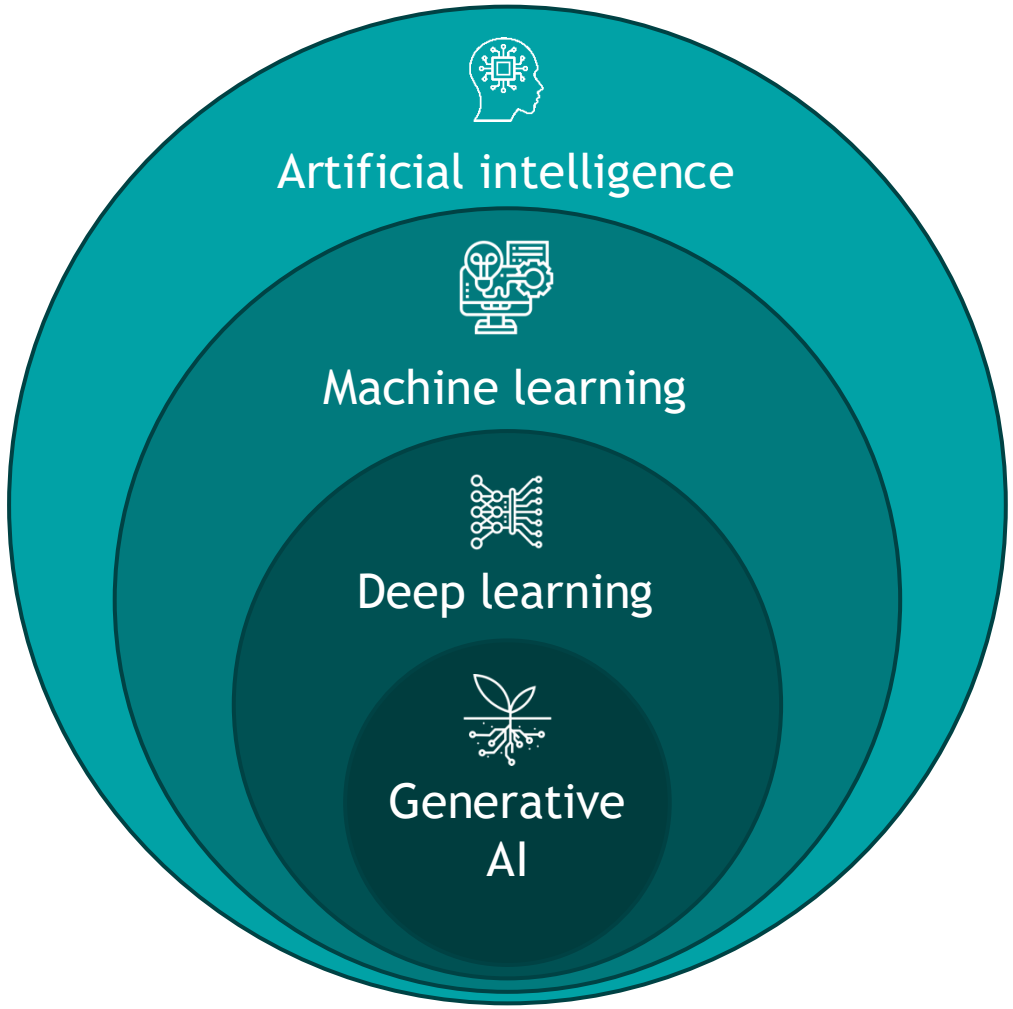


Artificial intelligence – machines show “human intelligence” (problem solving, decision-making, etc.)

e.g. DeepBlue (logic-based chess machine), Google Maps

Google Maps

Artificial intelligence



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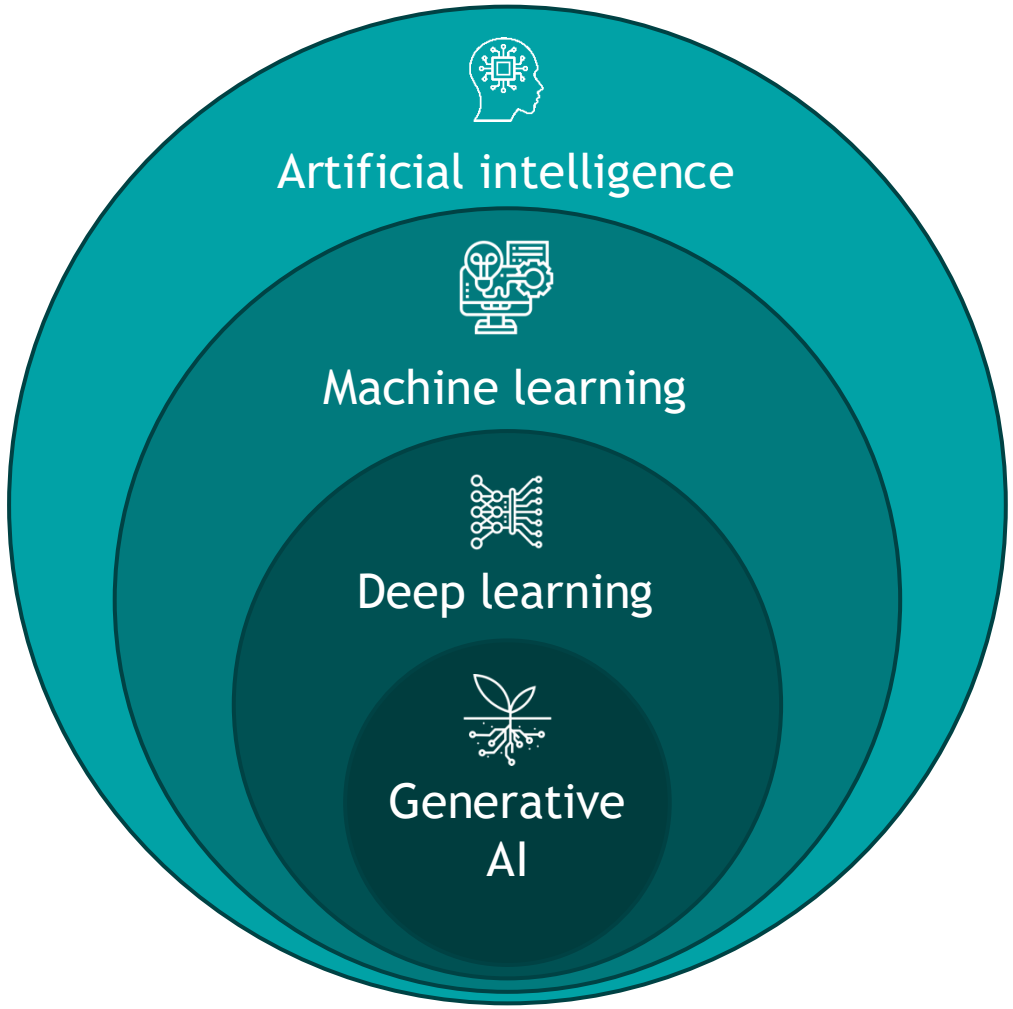

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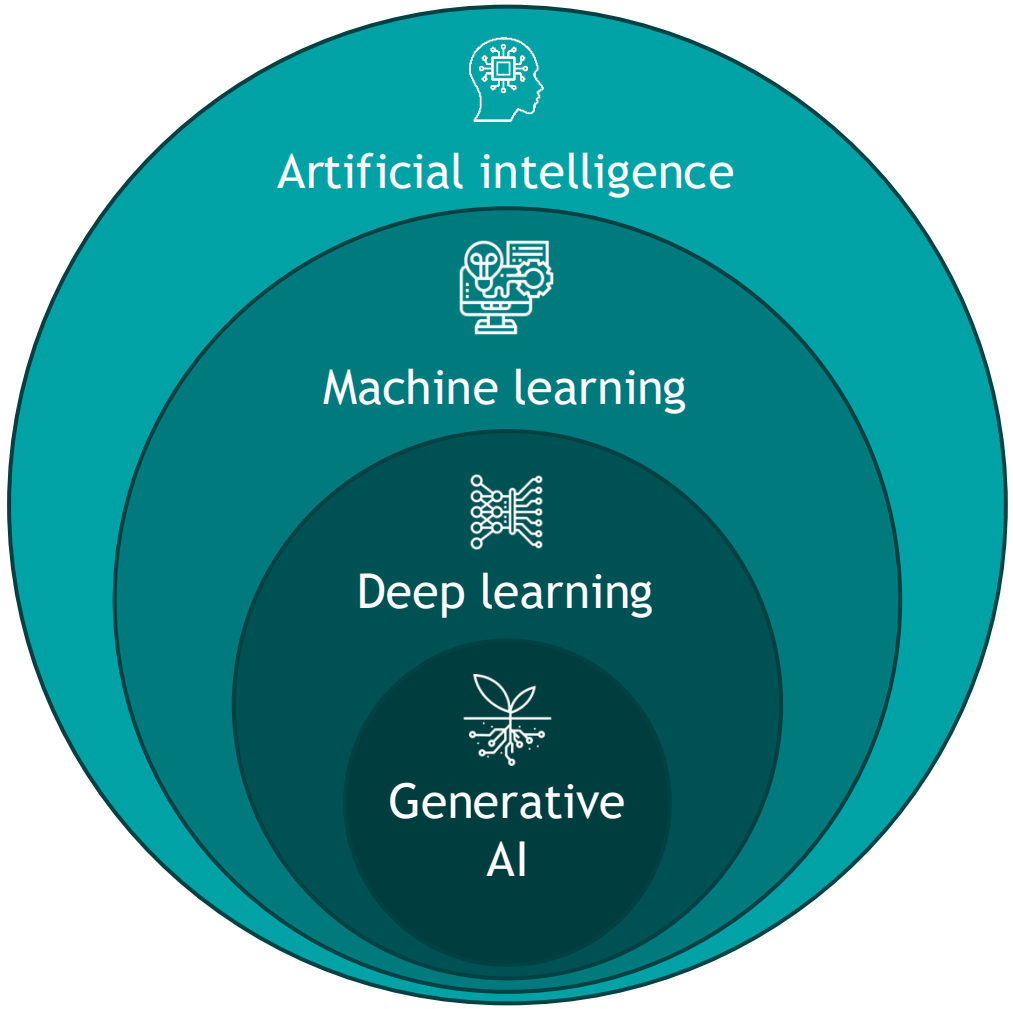
Deep learning – machine learning with deep neural networks

e.g. Face recognition, speech-to-text

Artificial intelligence



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Generative AI – creates new content based on learned patterns

e.g. ChatGPT

Gemini

What is the difference between machine learning and statistics?

What is machine learning?

THIS MONTH

POINTS OF SIGNIFICANCE

Statistics versus machine learning

Statistics draws population inferences from a sample, and machine learning finds generalizable predictive patterns.

Two major goals in the study of biological systems are inference and prediction. Inference creates a mathematical model of the data-generation process to formalize understanding or test a hypothesis about how the system behaves. Prediction aims at forecasting unobserved outcomes or future behavior, such as whether a mouse with a given gene expression pattern has a disease. Prediction makes it possible to identify best courses of action (e.g., treatment choice) without requiring understanding of the underlying mechanisms. In

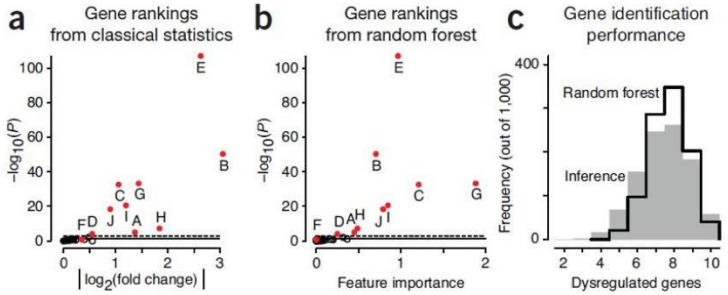


Figure 2 | Analysis of gene ranking by classical inference and ML. (**a**) Unadjusted log-scaled *P* values from statistical differential expression analysis as a function of effect size, measured by fold change in expression. (**b**) Log-scaled *P* values from **a** as a function of gene importance from random forest classification. In **a** and **b**, red circles identify the ten differentially expressed genes from **Figure 1**; the remaining genes are indicated by open circles. (**c**) Distribution of the number of dysregulated genes correctly identified in 1,000 simulations by inference (gray fill) and random forest (black line).

number of subjects, in contrast to 'long data', where the number of

Bzdok, D., Altman, N. & Krzywinski, M. Statistics versus machine learning. Nat Methods 15, 233–234 (2018)

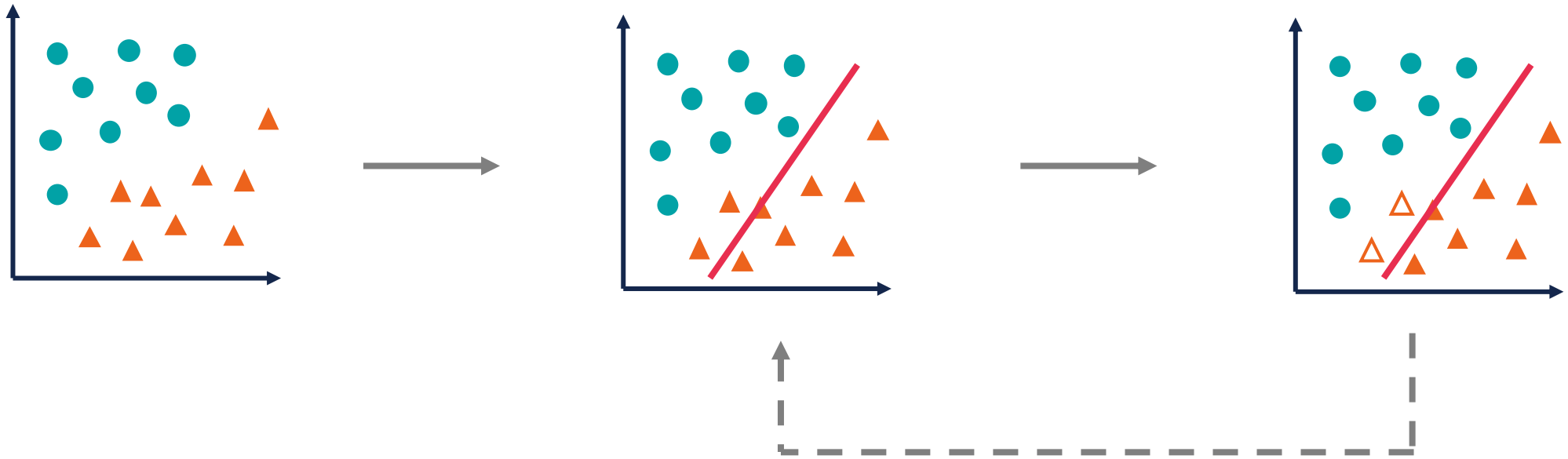
What is machine learning?

DATA

MODEL

Learns patterns
& decision rules

PREDICTION



VIB

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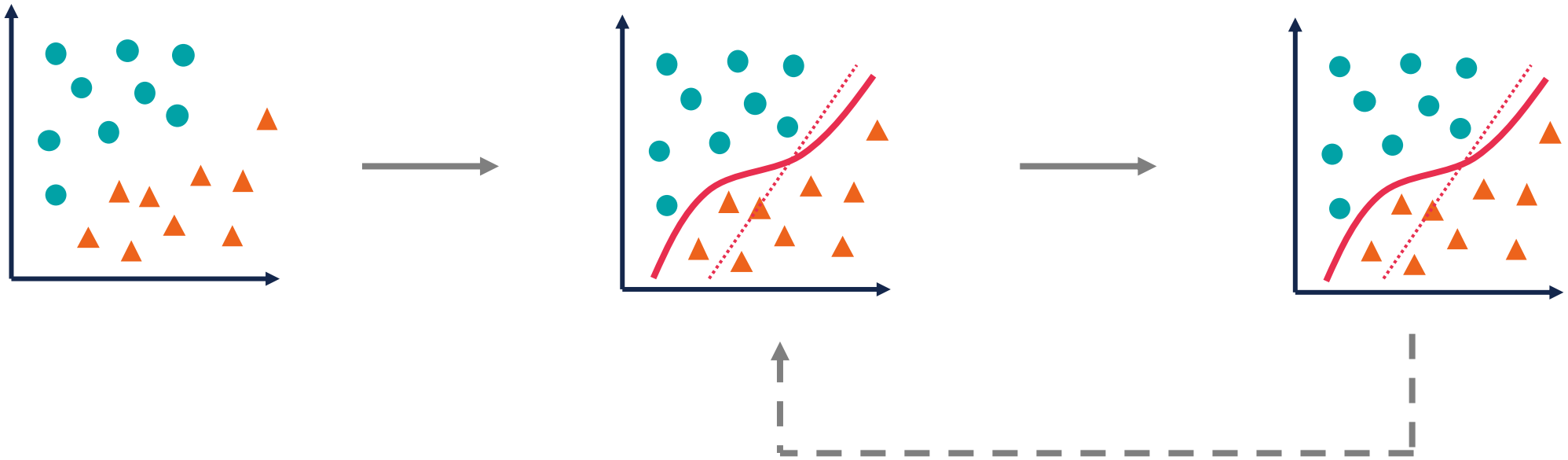
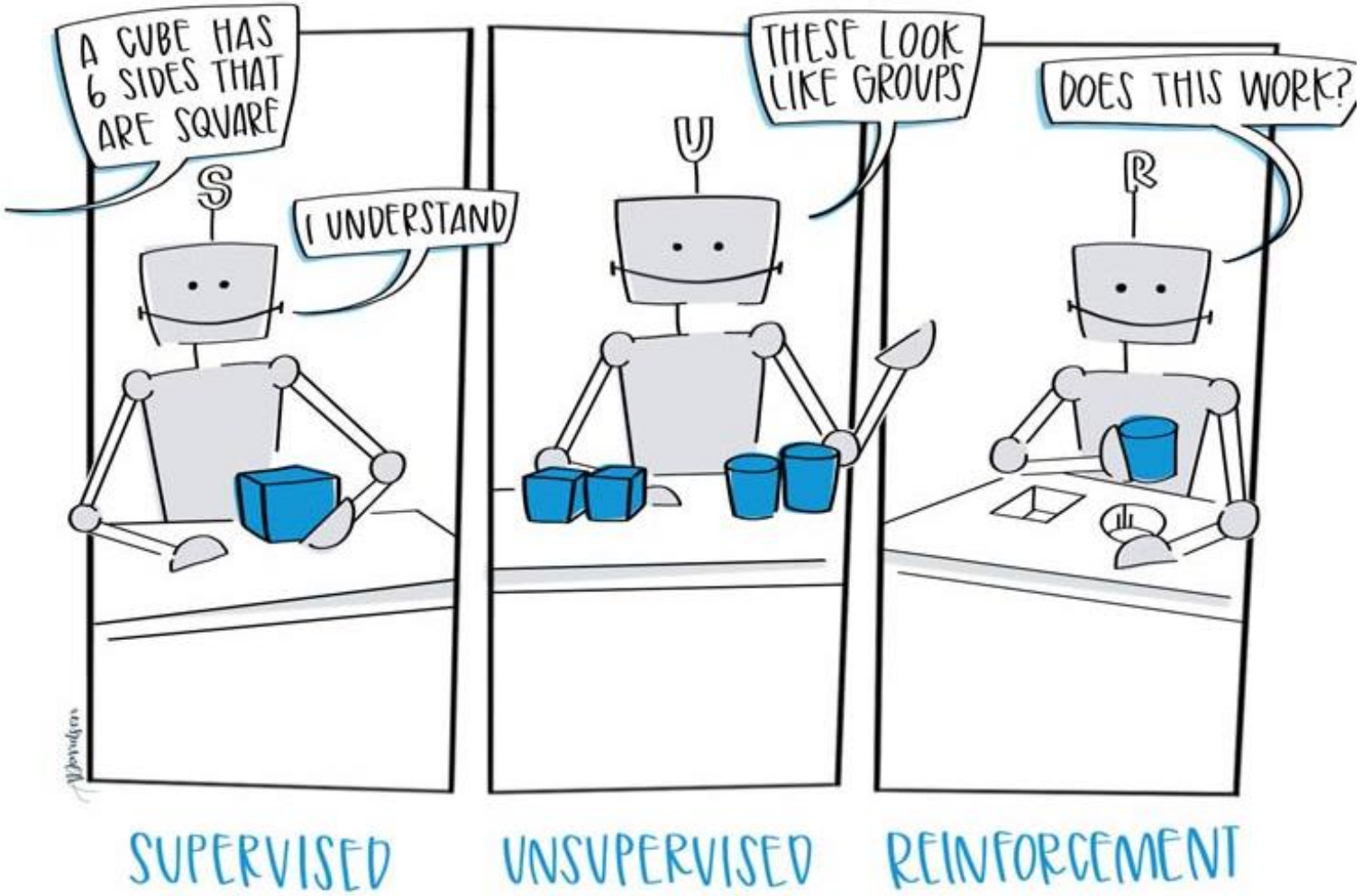


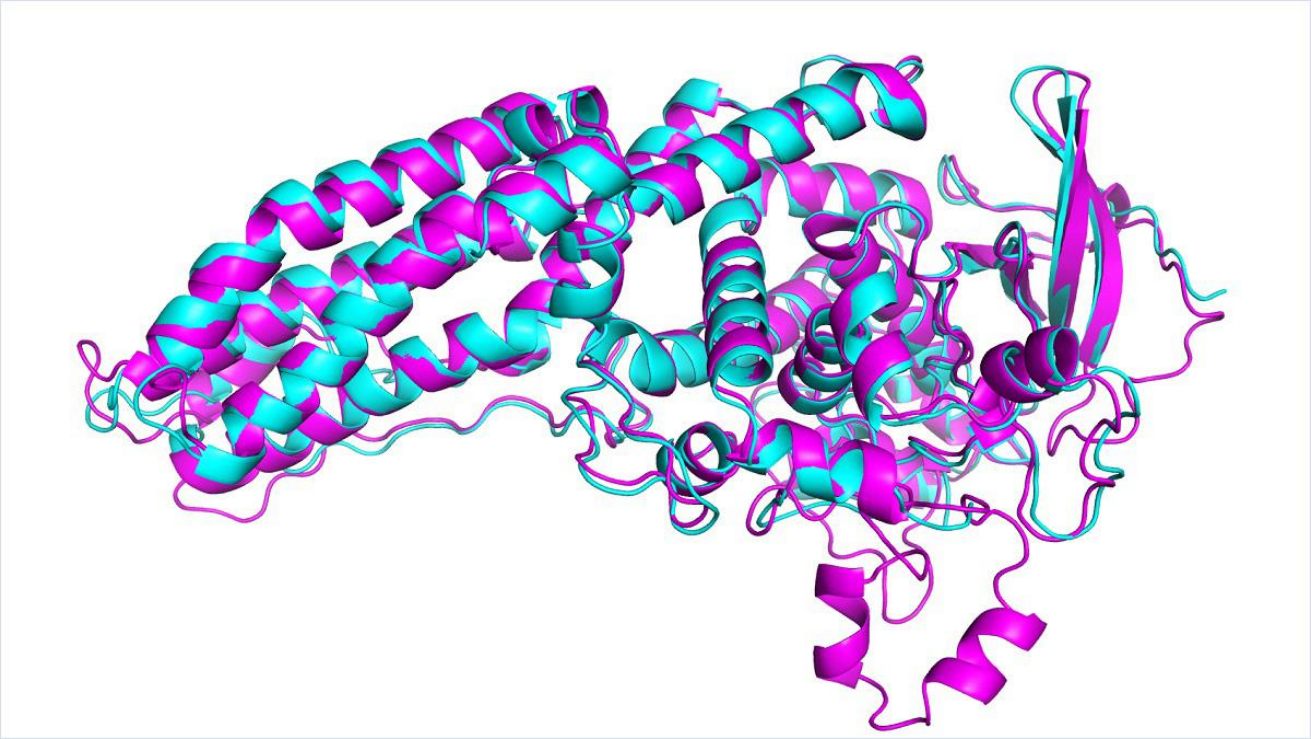
VIB



ceralytics

VIB

What is Machine Learning?



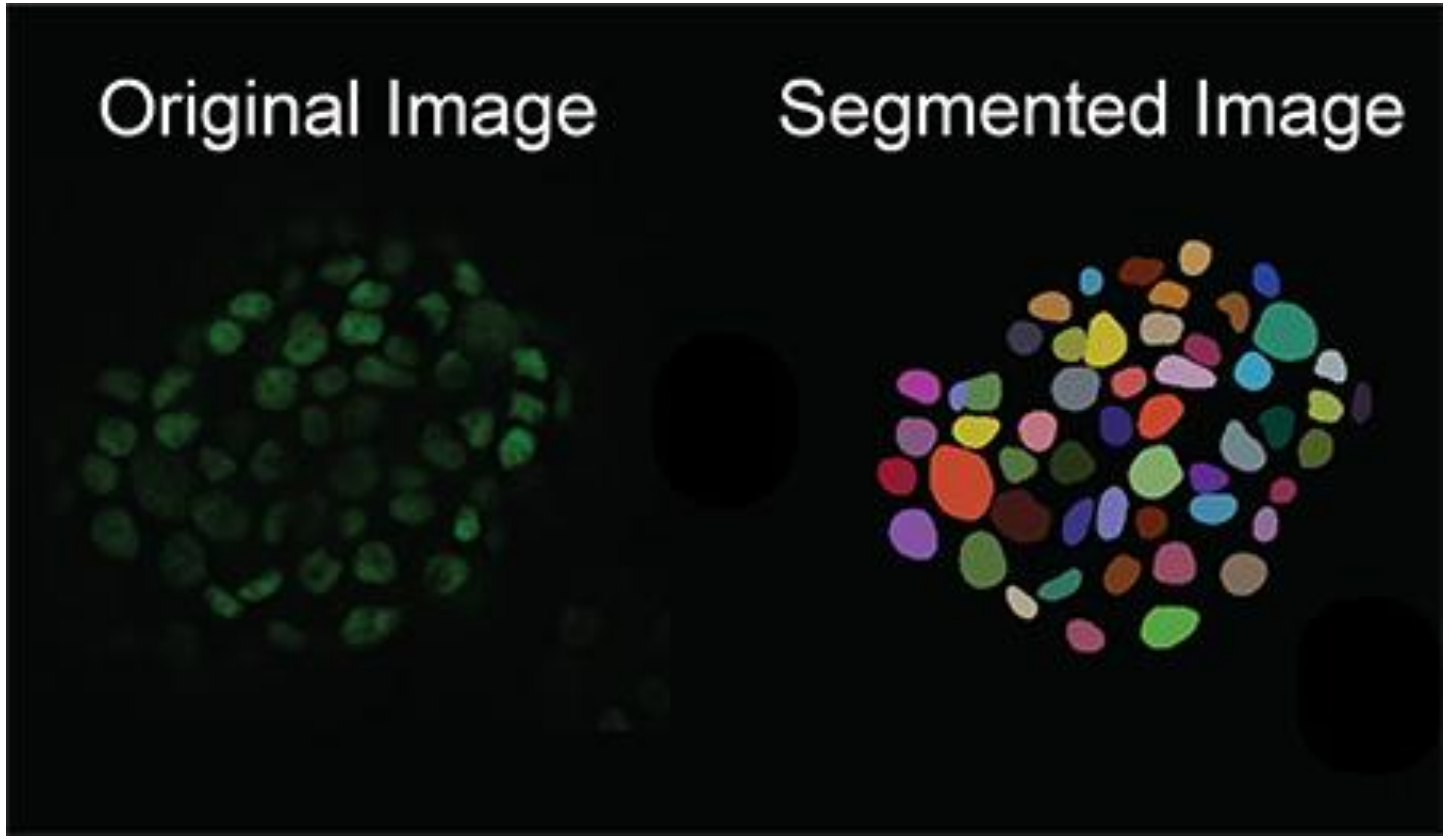
CASP/DEEPMIND

"Based on the amino acid sequence, can I predict the structure of the protein?"

VIB

What is Machine Learning?

”Based on the pixel values, can I predict where the cells are?”

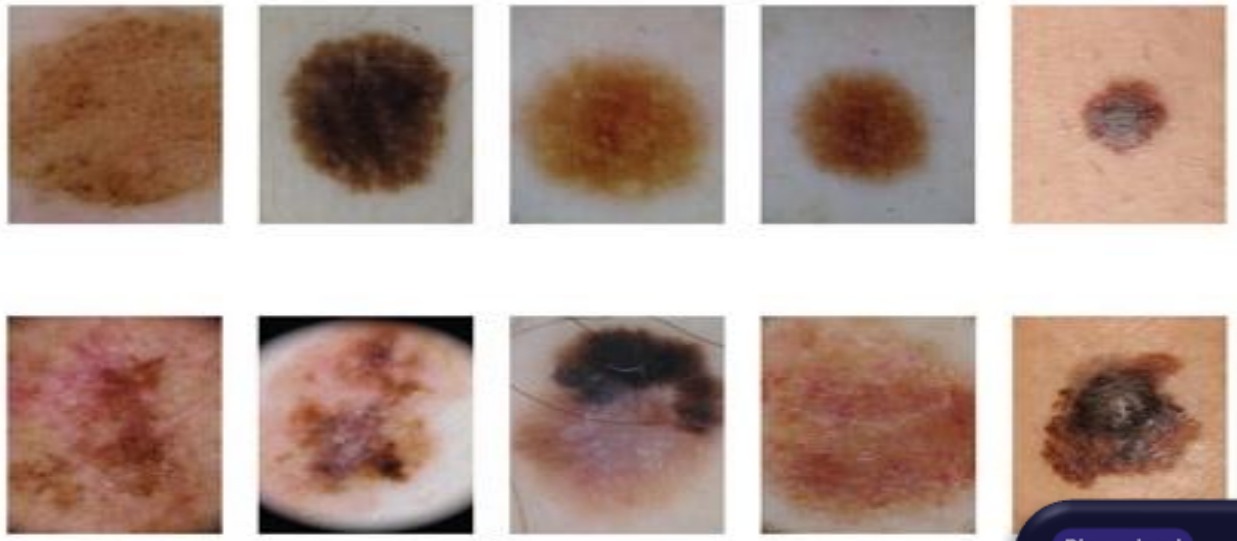


UC Santa Cruz

VIB

Melanoma detection

Assymetry **B**order **C**olor **D**iameter **E**volution

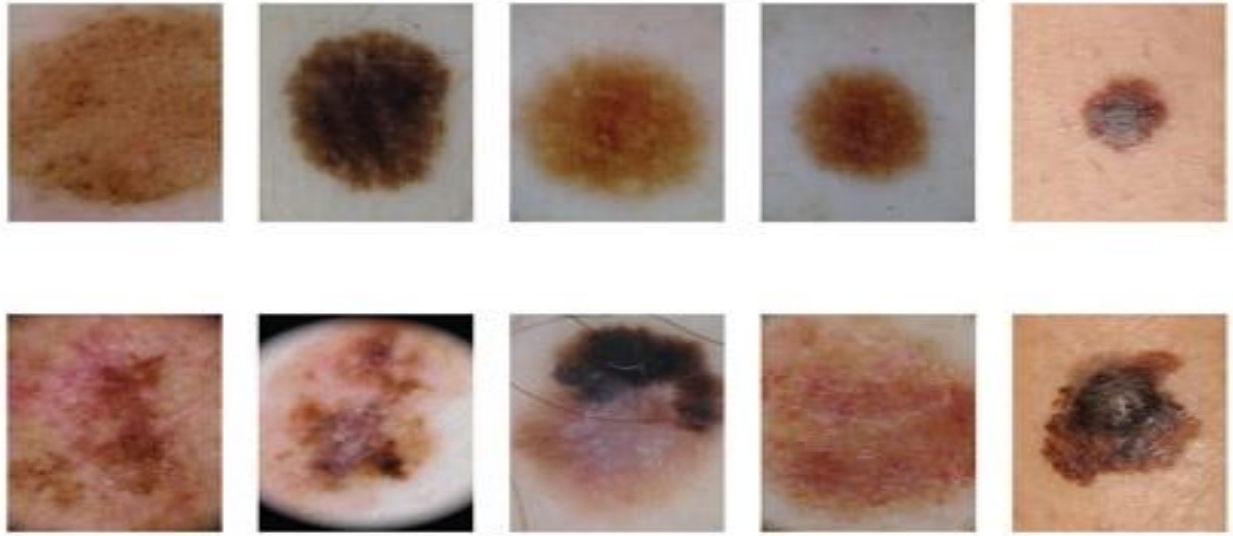


”Based on the lesion characteristics, can I predict its malignity?”

VIB

Machine learning - key concepts

Assymetry **B**order **C**olor **D**iameter **E**volution



Samples

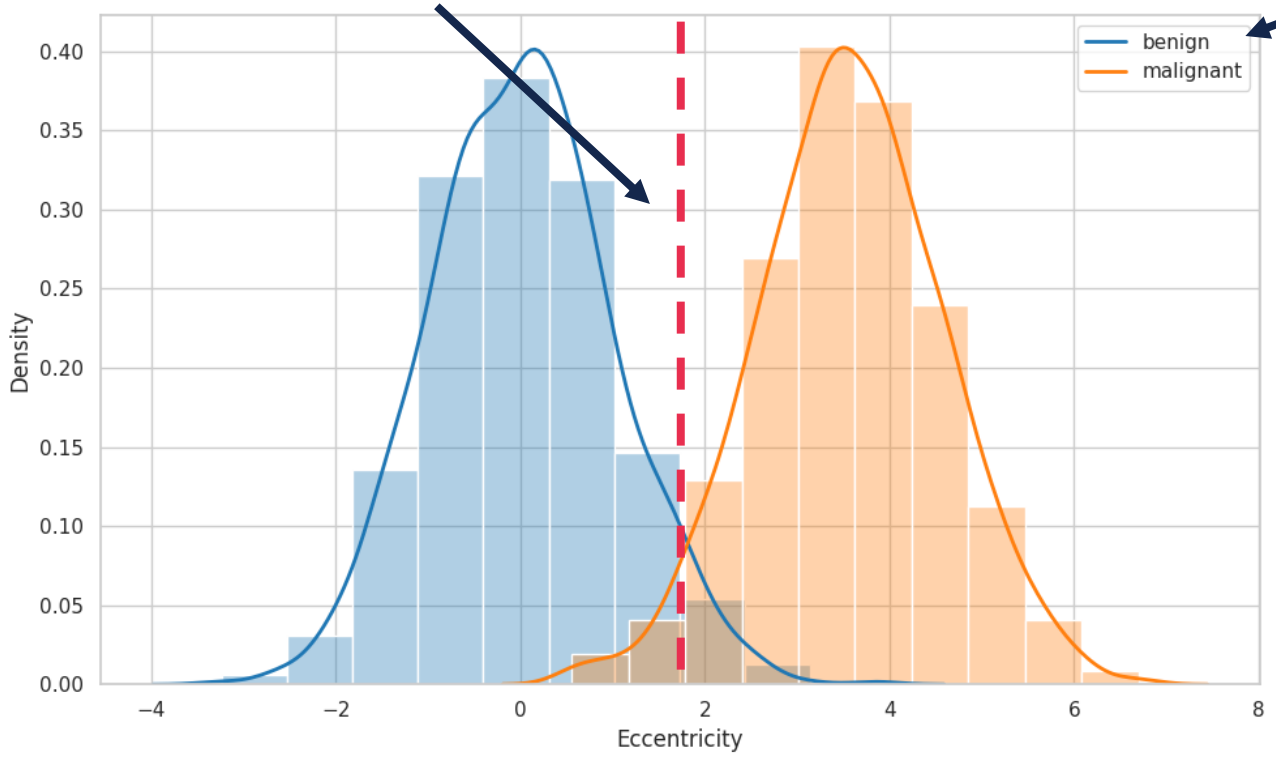
Features Labels

	A	B	C
1	Eccentricity	Solidity	label
2	-0.598720491	-2.863000525	0
3	-0.518610564	0.126108156	0
4	-1.843885879	-2.163015307	0
5	-0.289294284	-1.502927842	0
6	-1.177039495	-2.112436188	0
7	-0.934795367	-1.11164475	0
8	-1.598846606	-0.066942958	0
9	-1.455127299	0.972633286	0
10	-0.786144288	-1.096376283	0
11	-0.940576854	-1.869216938	0
12	-1.332590084	-0.206261961	0
13	-0.381720583	-0.413528409	1
14	-1.261911388	-0.946902992	0
15	-1.636395728	-1.201188514	0
16	-1.40571709	-1.605459274	0
17	-1.452295463	-1.266016012	0
18	-1.430260883	-1.378927623	0
19	-1.228654033	-2.1408459	0
20	-0.97102196	-2.366513251	0

Machine learning - key concepts

Model: threshold **t**

Labels: class of the examples



Features (characteristics computed from the data)

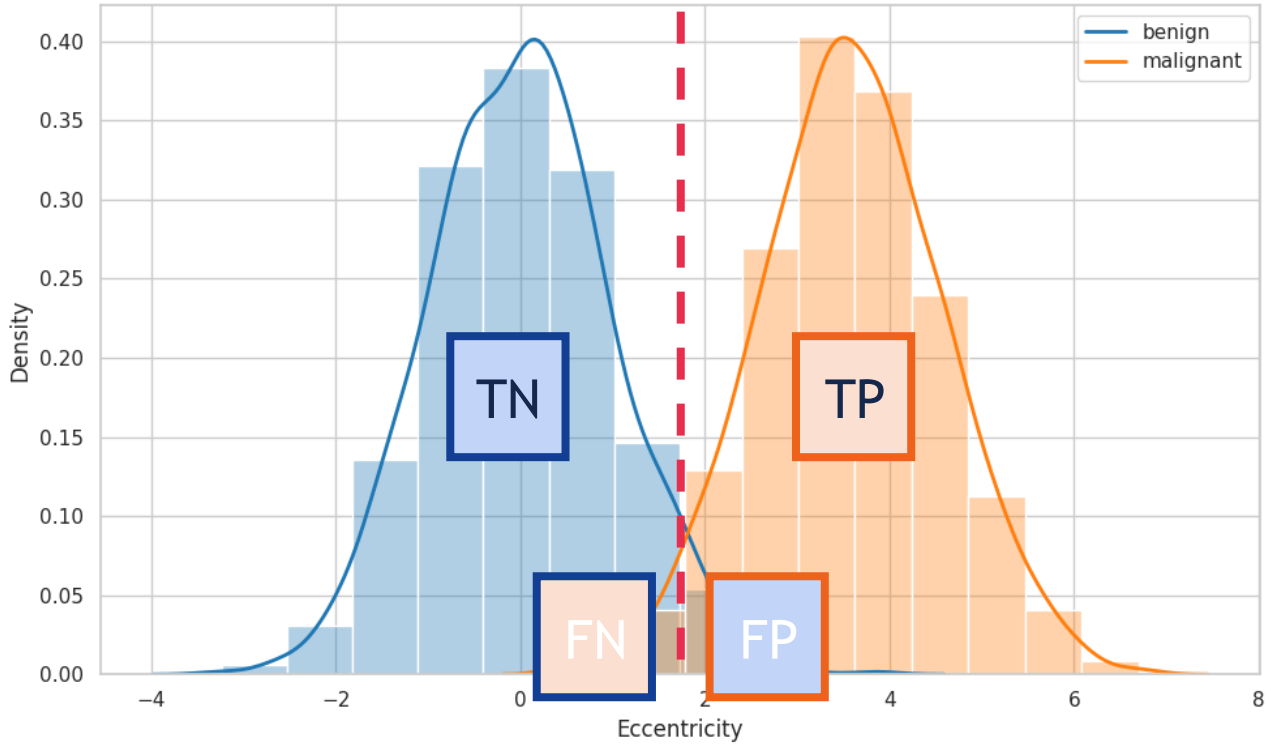
The **model** learns its prediction rules based on example data (**features** and **labels**)

For a new sample, the **features** are used to predict its **label**

How can we evaluate our model?

(supervised learning)

Evaluation metrics

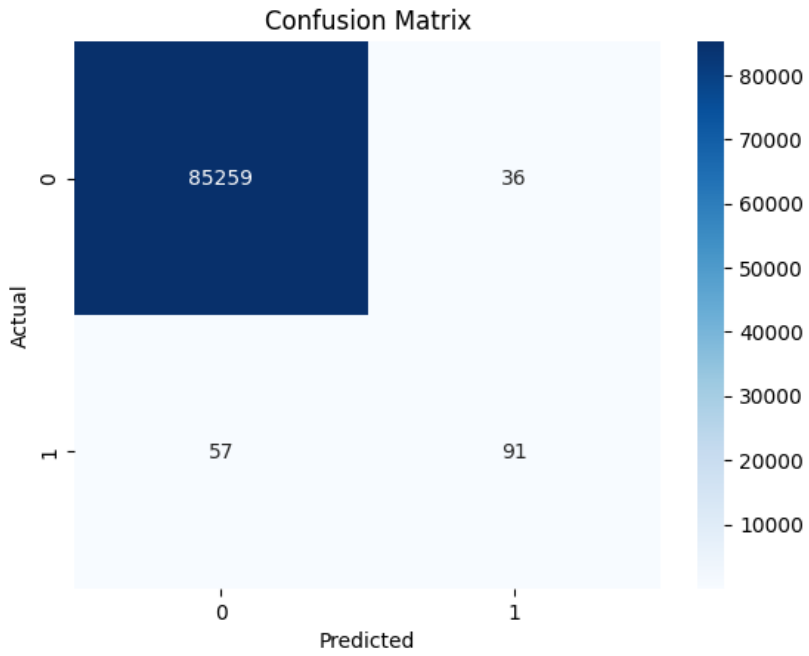


Confusion matrix

Classified as …

Really is …	*positive*	*negative*
positive	TP	FN
negative	FP	TN

Evaluation metrics

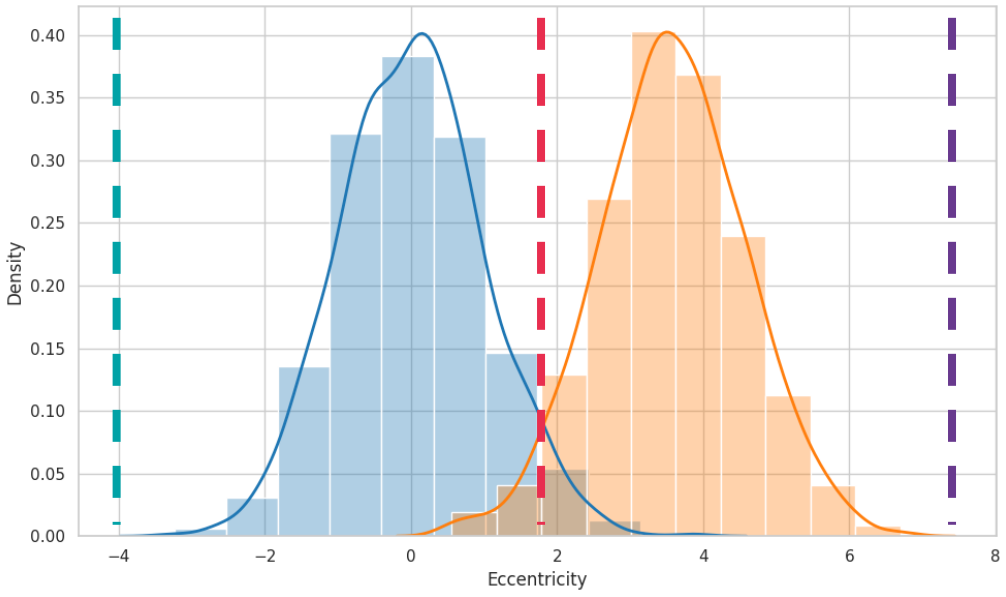


Metric	Formula
True positive rate, recall	$\frac{TP}{TP+FN}$
False positive rate	$\frac{FP}{FP+TN}$
Precision	$\frac{TP}{TP+FP}$
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$
F-measure	$\frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

Where should we put our focus? Which metric provides the best measure of performance?

F1-score ignores false negatives, and is robust against negative class imbalance

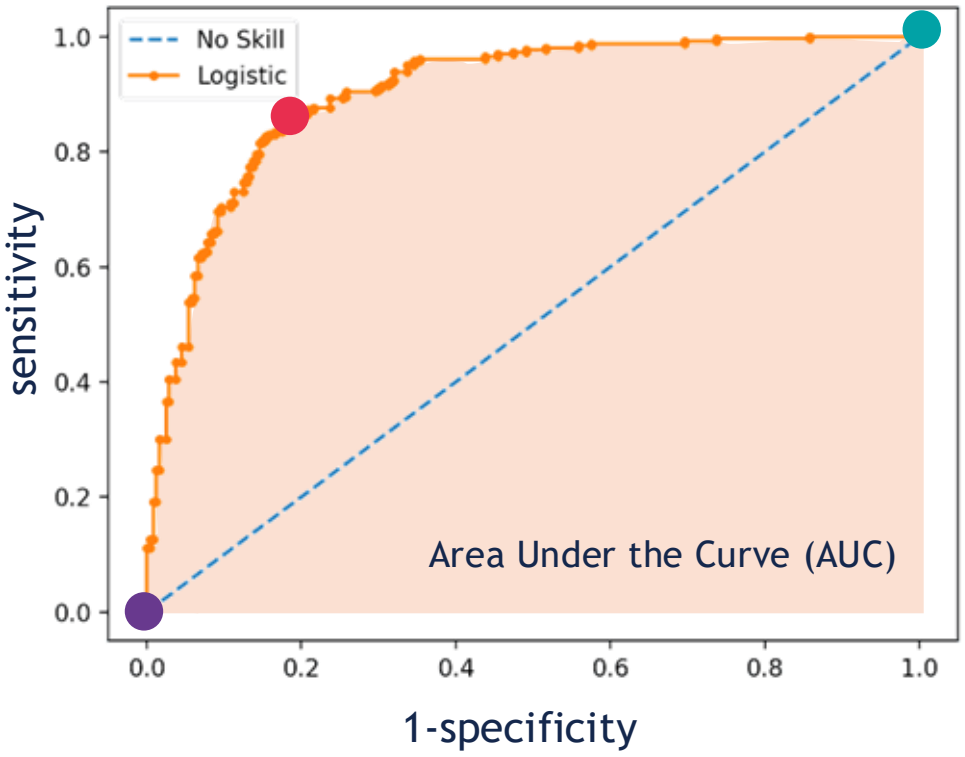
Evaluation metrics



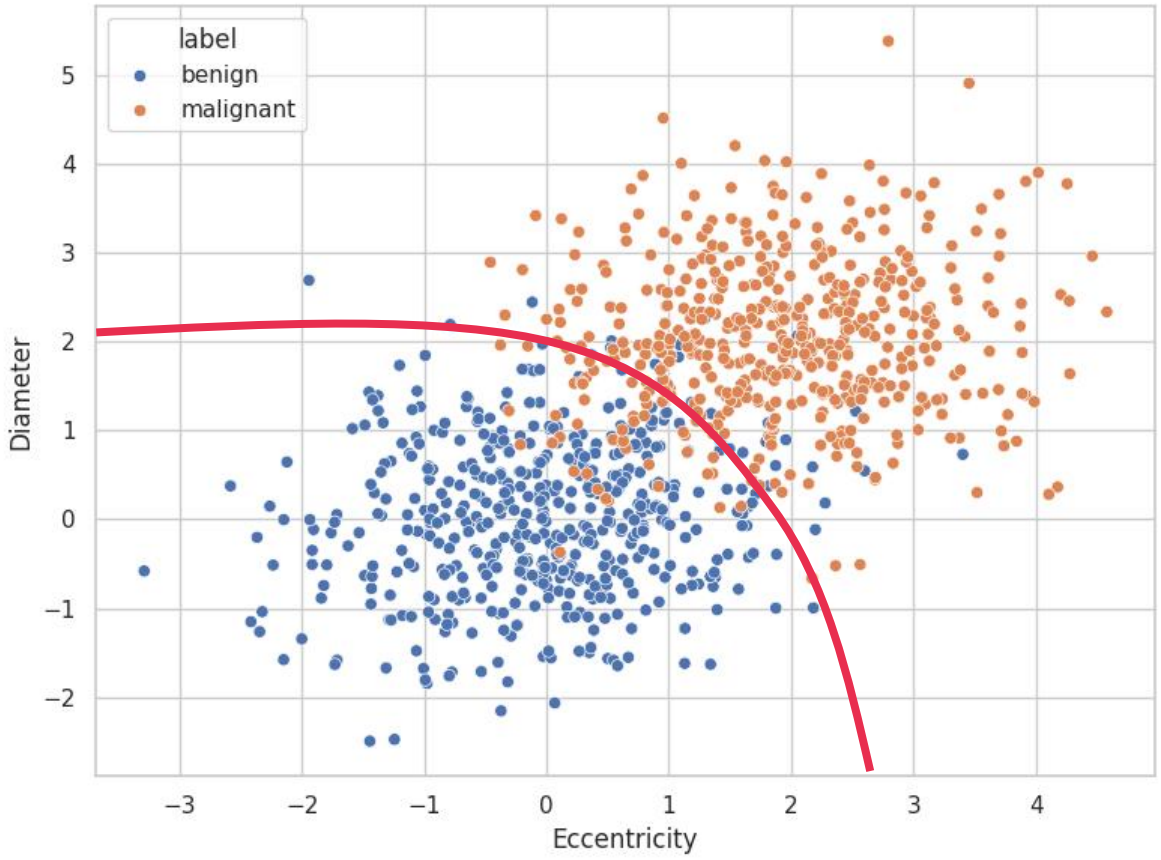
Feature: eccentricity of the lesion

Model: threshold **t**

Receiver Operating Characteristic (ROC) curve



Feature space

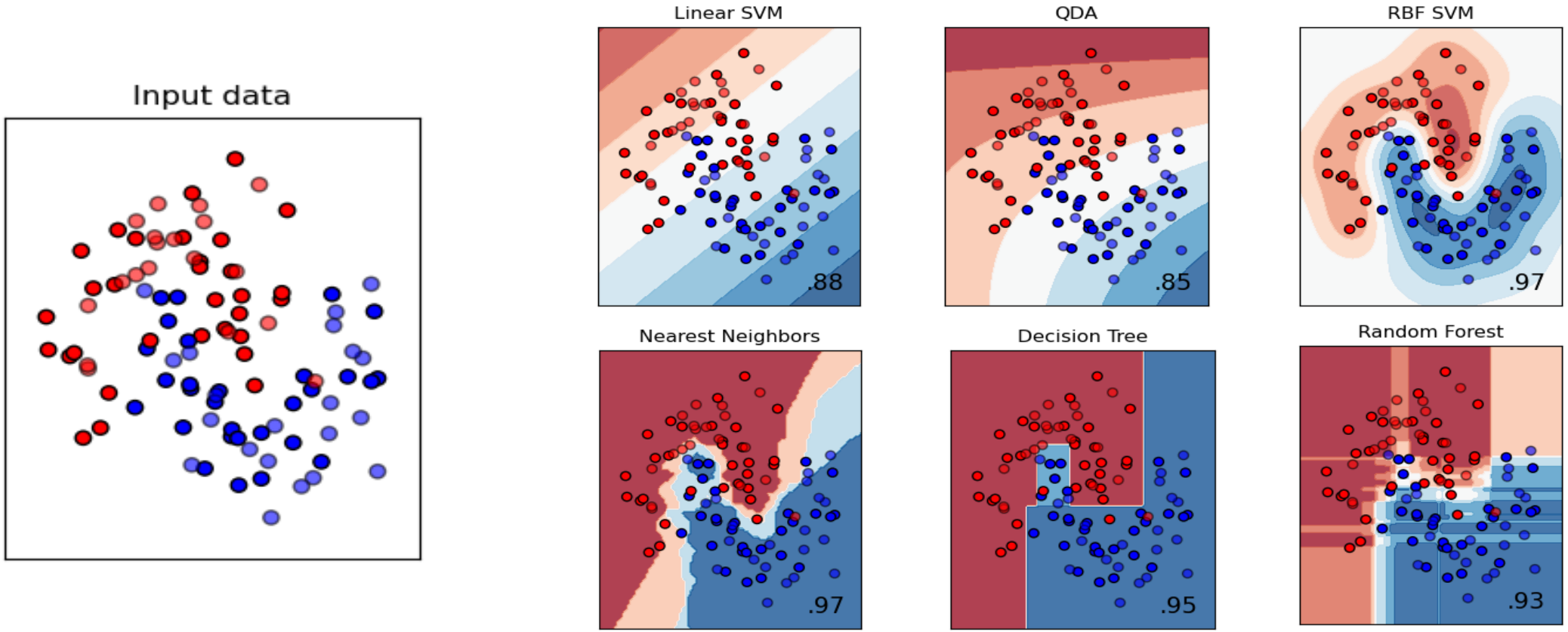


Combine multiple features to improve the predictive capabilities

→ Feature vector X

→ Euclidean vector space

Model complexity



Data complexity ↑ → Model complexity ↑ → # of examples needed ↑